# Signal theory for images

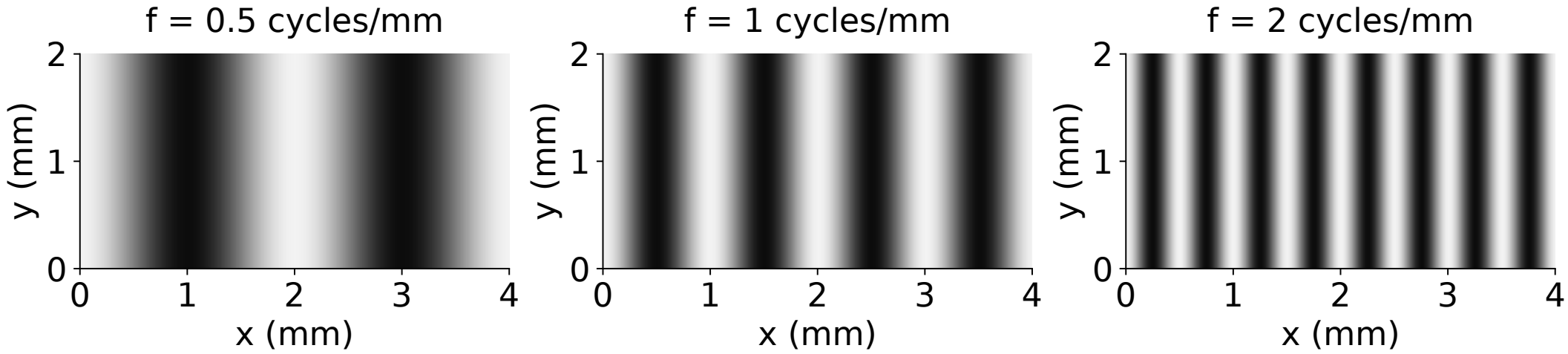
# Spatial signals

An image can be represented by a function $I(x, y)$.

A line through the image gives a one-dimensional intensity profile, $I(x, y_0)$.

We can apply the same sinusoidal and Fourier descriptions used for time signals, with position replacing time.

# A sinusoidal resolution target




One bright-dark pair is one cycle. Increasing spatial frequency brings the stripes closer together.

# Mean, amplitude, and phase

$$I(x) = \overline{I} + A\cos(2\pi f x + \phi).$$

| Symbol | Meaning | Units in this exercise |
|---|---|---|
| $\overline{I}$ | Mean intensity | Normalized intensity |
| $A$ | Sinusoidal amplitude | Normalized intensity |
| $f$ | Spatial frequency | Cycles/mm |
| $\phi$ | Phase | Radians in the equation |

The notebook uses $\overline{I} = 0.5$ and $A = 0.5C_0$.

# Stripe period

The period is the distance needed for one complete cycle:

$$T_x = \frac{1}{f}.$$

For $f = 2$ cycles/mm, $T_x = 0.50$ mm.

**Check:** a pattern has three bright-dark pairs across 2 mm. What are its spatial frequency and period?

# Spatial period and frequency conventions

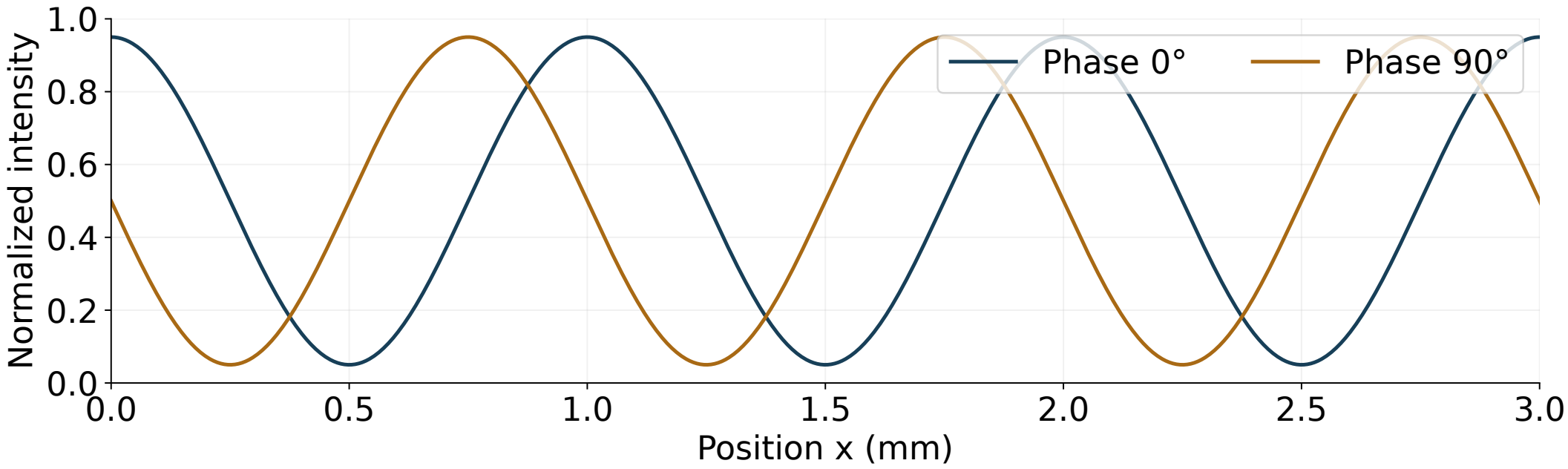
The spatial period $T_x$ can also be written as $\Lambda_x$, uppercase lambda.

$$I(x) = \overline{I} + A\cos(2\pi u_0 x + \phi) = \overline{I} + A\cos(k_{x0}x + \phi).$$

$$u_0 = \frac{1}{\Lambda_x} \quad \text{(cycles/mm)}, \qquad k_{x0} = 2\pi u_0 = \frac{2\pi}{\Lambda_x} \quad \text{(rad/mm)}.$$

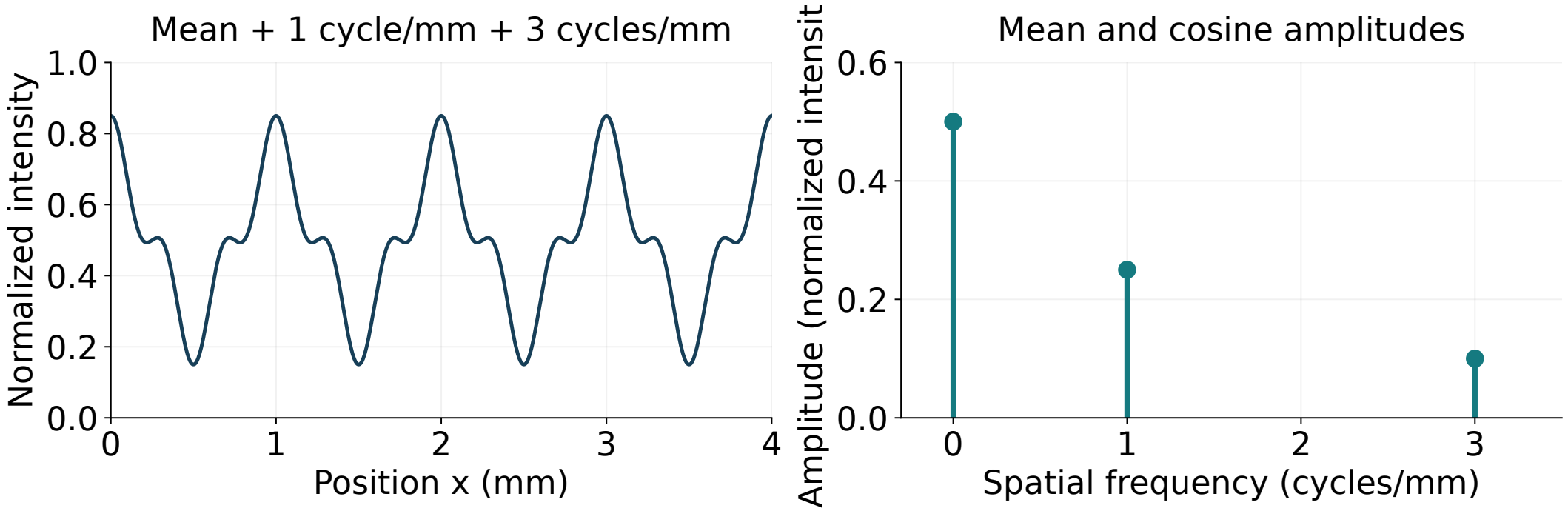
$x$ is a position; $\Lambda_x$ is a repeating distance. The notebook's frequency $f$ is a cycles-per-length frequency, like $u_0$.

# Phase shifts the pattern




Both signals have the same frequency and amplitude.

# A signal as a sum of spatial frequencies




The mean, or zero-frequency component, sets the overall intensity level.

# Spatial Fourier transform

For a one-dimensional profile, we use

$$\tilde{I}(f) = \int_{-\infty}^{\infty} I(x) e^{-j2\pi f x}\, dx.$$

- $x$ is position in mm and $f$ is spatial frequency in cycles/mm.
- Magnitude describes the strength of each frequency component.
- Phase describes its spatial alignment.

The inverse transform uses $e^{+j2\pi f x}$.

# Two-dimensional Fourier coordinates

Positions $(x, y)$ correspond to spatial frequencies $(u, v)$:

$$\tilde{I}(u, v) = \iint I(x, y) e^{-j2\pi(ux+vy)}\, dx\, dy.$$

With angular spatial frequencies,

$$\hat{I}(k_x, k_y) = \iint I(x, y) e^{-j(k_x x + k_y y)}\, dx\, dy.$$

$$k_x = 2\pi u, \qquad k_y = 2\pi v.$$

If $x, y$ are in mm, then $u, v$ are in cycles/mm and $k_x, k_y$ are in rad/mm.

# Magnitude and phase of a complex spectrum

Write $F(u,v) = F_R(u,v) + jF_I(u,v)$, where $j^2 = -1$.

$$|F| = \sqrt{F_R^2 + F_I^2}, \qquad \angle F = \text{atan2}(F_I, F_R).$$

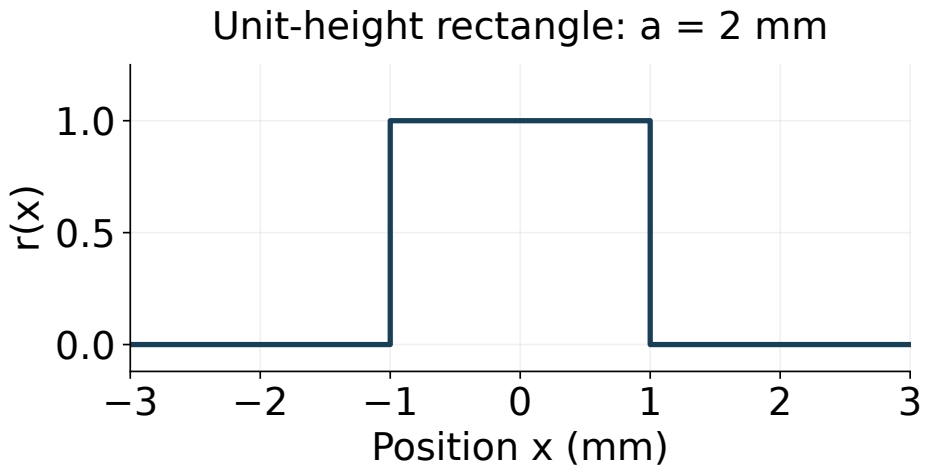
The two-argument angle function identifies the correct quadrant. Phase is undefined where $|F| = 0$.

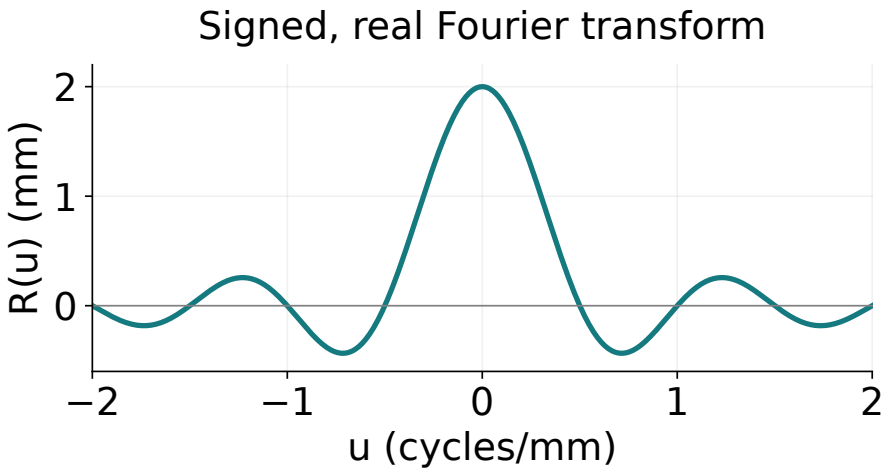
**Example:** for $F = -1 + j\sqrt{3}$, the magnitude is 2 and the phase is $2\pi/3$ radians, or $120°$.

# A rectangular spatial signal and its transform

A unit-height rectangle $r(x) = \text{rect}(x/a)$ equals 1 for $|x| < a/2$ and 0 outside. Its width is $a > 0$.

$$R(u) = a\ \text{sinc}(au), \qquad \text{sinc}(z) = \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}(0) = 1.$$

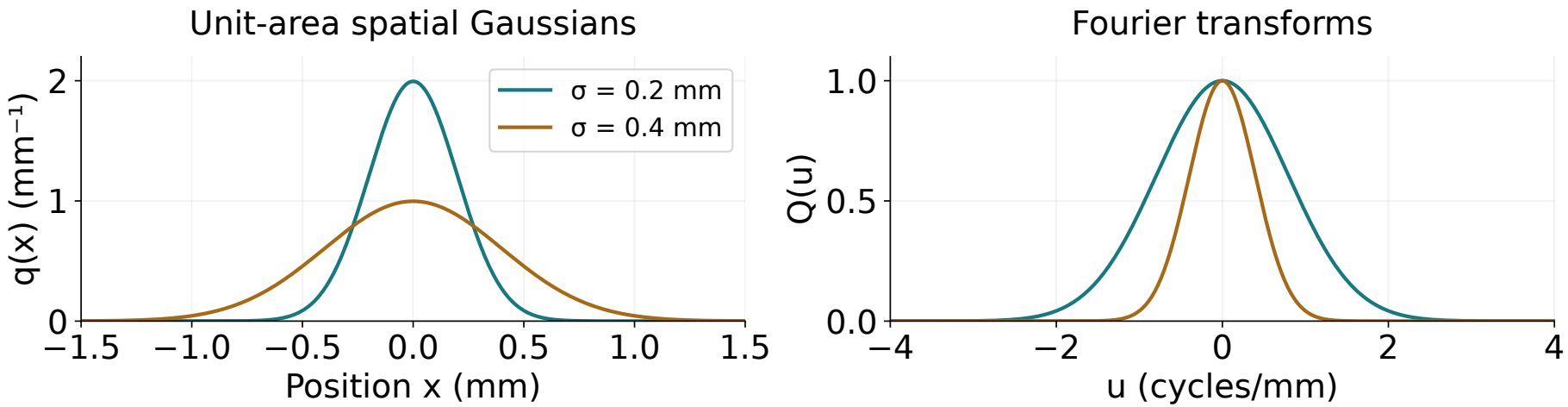


# A Gaussian spatial signal and its transform

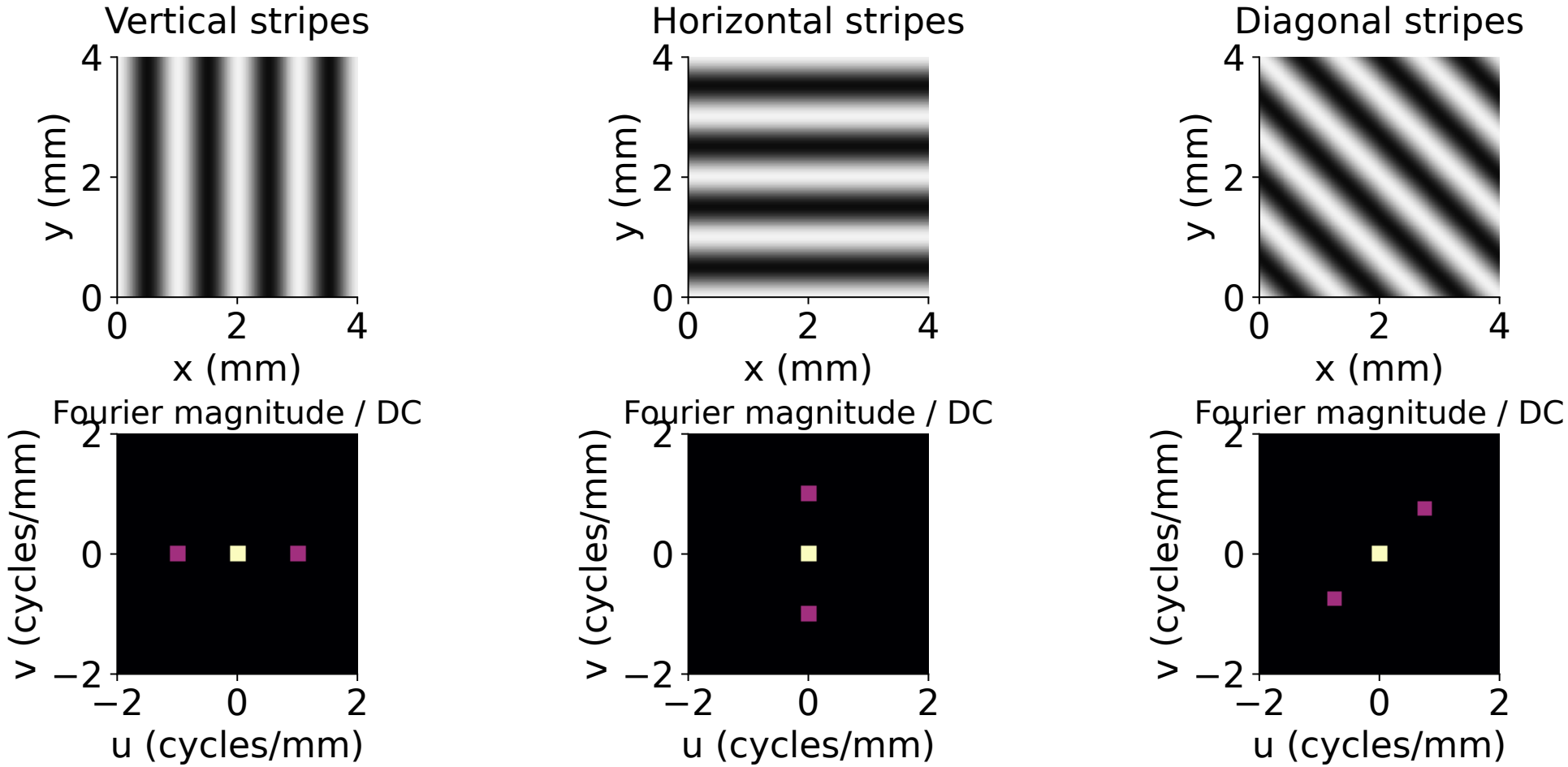
For a centered Gaussian with unit area,

$$q(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}, \qquad Q(u) = e^{-2\pi^2\sigma^2u^2}.$$




A wider spatial Gaussian has a narrower frequency response.

# Frequency has a direction in two dimensions




The frequency vector points in the direction of intensity variation, perpendicular to the stripes.

# Spatial sampling

For ideal point samples with spacing $p$,

$$I_n = I(np), \qquad f_s = \frac{1}{p}.$$

The sampling frequency $f_s$ is the number of samples per mm.

For a sinusoid, the number of samples per period is

$$N_{\text{period}} = \frac{1}{pf}.$$

# Nyquist frequency

$$f_N = \frac{f_s}{2} = \frac{1}{2p}.$$

For an ideal band-limited signal, recovery from equally spaced samples requires its highest frequency to be strictly below $f_N$.

A sinusoid below this limit has more than two samples per period.

**Example:** $p = 0.25$ mm gives $f_s = 4$ samples/mm and $f_N = 2$ cycles/mm.

# A sampling limit in angular-frequency units

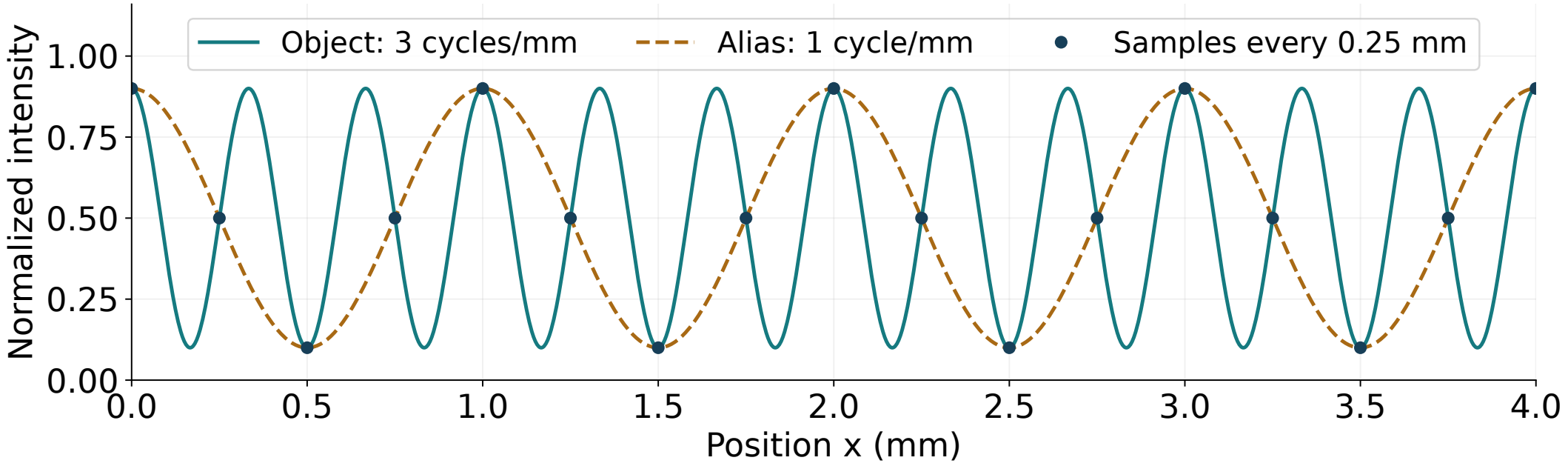
For pixel spacing $p$, the angular sampling frequency is $k_s = 2\pi/p$ and the angular Nyquist frequency is $k_N = \pi/p$.

$$k_{\text{max}} < \frac{\pi}{p}, \qquad p < \frac{\pi}{k_{\text{max}}}.$$

**Example:** if $k_{\text{max}} = 10\pi$ rad/mm, then $u_{\text{max}} = 5$ cycles/mm and the limiting pitch is 0.10 mm.

Choose a smaller pitch for phase-independent sampling of the highest-frequency component. Exactly at the limit there are only two samples per period.

# Aliasing




At these sample locations, the two continuous signals give identical values.

# A sampling calculation

A detector records a 1.5 cycles/mm stripe pattern.

| Quantity | $p = 0.20$ mm | $p = 0.40$ mm |
|---|---|---|
| Sampling frequency $f_s$ | ? | ? |
| Nyquist frequency $f_N$ | ? | ? |
| Samples per stripe period | ? | ? |
| Target below Nyquist? | ? | ? |

Calculate the entries with a partner. Predict whether either detector can show a false stripe spacing.

# Sampling and physical blur

An object can contain frequencies above a detector's Nyquist frequency.

Physical blur reduces high-frequency contrast before sampling. This can weaken aliases, while also reducing useful fine detail.

Increasing the displayed image size adds no new detector measurements.

Next meeting: the point-spread function and modulation transfer function quantify these effects.

# Exit questions

1. What changes when spatial frequency doubles: stripe period, amplitude, or both?
2. What is the Nyquist frequency for $p = 0.10$ mm?
3. Why can two different continuous patterns agree at every recorded sample?

# Reading and resources

Prince and Links, *Medical Imaging: Signals and Systems*: spatial signals, Fourier representations, and sampling.

- MIT RES.6-007: lecture notes on convolution, Fourier transforms, sampling, and interpolation.
- MIT HST.582J: lecture notes for further biomedical signal examples.